“Amaze your friends! Baffle your enemies!”

And the story continues.

A long time ago (shortly after I finished college in 1977) I got my first computer, a TRS-80 model 1. In the early days of personal computing, many computer peripherals, such as printers and floppy disk drives, were very costly and as a result, my dad (an electrical engineer) and I would cruise electronic surplus stores looking for deals on devices we could attach to our new computer.

One day, as we were searching a large warehouse near the University of Maryland, I came across a store display featuring a small, clear, plastic box containing a battery, a few computer chips, and several randomly blinking LEDs. While the little device served no useful purpose, it did have blinking lights. Above it hung a handwritten sign that read simply:

Amaze Your Friends! Baffle Your Enemies!

The excitement pervasive in the early days of personal computing is hard to explain to people today. The computers of that period seem so laughably primitive by today’s standards but it was a revolution nonetheless and there were many explorers mapping the new, uncharted territory of personal empowerment and technical innovation.

People entering the computer field now are at a disadvantage compared to those of us who came up in the 1970s and 1980s. The early computers were very simple, slow, and had tiny memories. All the attributes you need if you really want to understand how computers work. Today, computers are so fast, and software so large and complex that you can’t see the computer underneath anymore and that’s a shame. You can’t see the beauty of what they do.

However, we are now in the midst of another revolution, extremely low-cost computing. Devices like the Raspberry Pi single board computer offer the opportunity to work on systems more simple and basic compared to contemporary desktop and mobile devices. But make no mistake, these low-cost computers are powerful. In fact, a $35 Raspberry Pi compares favorably to the $30,000 Unix workstations I used in the early 1990s.

What this book is about

This volume is a sequel/supplement to my first book, *The Linux Command Line* (TLCL) and as such, we will be referring back to the first book frequently, so if you don’t already have a copy, please download one from LinuxCommand.org or, if you prefer, pickup a printed copy from your favorite bookseller or library. This time around we are going to

build on our experience with the command line and add some more tools and techniques to our repertoire. Like the first book, this second volume is not a book about Linux system administration, rather it is a collection of topics that I consider both fun and interesting. We will cover many tools that will be of interest to budding system administrators, but the tools were chosen for other reasons. Sometimes they were chosen because they are “classic” Unix, others because they are just “something you should know,” but mostly topics were chosen because I find them fun and interesting. Personal computing, after all, should be about doing things that are fun and interesting just as it was in the early days.

Who should read this book

This is a book for explorers and creators looking for adventure. I think computers are the coolest things ever and if you share that feeling of excitement with every new thing you can get your computer to do then you have come to the right place. Many people today come into the computer field only in hopes of developing enough skill to get a job. There is nothing wrong with that of course. Everyone needs to earn a decent living, but there is more to life than that. There is beauty and there is love, and if you are wise (and lucky) you will find these things in your career. Computers are powerful tools that, in the right hands, can improve the human condition. I think it’s a worthy goal to leave the world a little better than the way you found it. I hope you do too.

What’s in the first Internet edition

For the most part, you can think of this book as an expansion of Part 3 of TLCL. In fact, I considered some of these topics for inclusion in the first book, but ran out of space for them. That being said, this is definitely a work-in-progress. Future editions will contain more chapters and the existing chapters will contain additional content and the chapters will likely appear in a different order. Typography and layout will improve too.

How to read this book

This book is not as linear as TLCL so feel free to skip around. Some adventures are prerequisites for later ones. If an adventure requires an earlier one, it will be indicated. A few of the adventures call for supplemental material (typically code samples and datasets) that can be downloaded from LinuxCommand.org.

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And as always, use your powers only for good. Let the adventures begin!